CIE International A Level Physics (9702)

Topics 14, 15 & 16
Thermal Physics

Temperature · Ideal Gases · Thermodynamics

Revision Booklet

Syllabus 9702 — A Level (A2) — 2025–2027

This booklet covers:

- Thermal Equilibrium and Temperature Scales
- Specific Heat Capacity and Specific Latent Heat
- The Mole and the Ideal Gas Equation
- Kinetic Theory of Gases
- Internal Energy
- The First Law of Thermodynamics
- Formula Summary, Worked Examples, Exam Practice & Checklist

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1 Core Concepts and Definitions

1.1 What is Temperature?

Temperature is a measure of the **average random kinetic energy** of the molecules in a substance. It is *not* the same as thermal (internal) energy: a large cold object may contain more internal energy than a small hot object, yet have a lower temperature.

Definition: Thermal Equilibrium

Two objects are in **thermal equilibrium** when there is no net transfer of energy between them — they are at the same temperature.

This is the basis of the *zeroth law of thermodynamics*: if objects A and B are each in thermal equilibrium with object C, then A and B are also in thermal equilibrium with each other.

2 Topic 14 — Temperature

2.1 Temperature Scales

The Celsius Scale

The Celsius scale is defined by two fixed points:

- 0 °C — the ice point (pure water–ice equilibrium at 1 atm)
- 100 °C — the steam point (pure water–steam equilibrium at 1 atm)

The Thermodynamic (Kelvin) Scale

The thermodynamic scale does **not** depend on the properties of any particular substance. Its zero point, **absolute zero**, is the temperature at which molecules have the minimum possible kinetic energy. No substance can be cooled below this point.

Temperature Conversion

$$T\,/\,\mathrm{K} = \theta\,/\,{}^\circ\mathrm{C} + 273.15 \tag{1}$$

T = thermodynamic temperature (K)
θ = Celsius temperature (°C)

✓ Exam Tip

Always convert temperatures to kelvin before substituting into gas-law equations. A temperature *difference* $\Delta\theta$ in °C equals the same numerical difference ΔT in K, so conversion is not needed for differences — but is always required for absolute values in $pV = nRT$.

2.2 Thermometric Properties

A **thermometric property** is a physical quantity that varies measurably and reproducibly with temperature, and can therefore be used to measure it.

Thermometric property	Type of thermometer
Volume of a gas at constant pressure	Gas thermometer
Resistance of a metal (e.g. platinum)	Resistance thermometer (PRT)
E.m.f. of a thermocouple	Thermocouple thermometer
Density of a liquid	Liquid-in-glass thermometer

Different thermometers may give slightly different readings between the fixed points because their thermometric properties do not all vary linearly with temperature in identical ways. The thermodynamic scale avoids this problem entirely.

2.3 Specific Heat Capacity

Definition: Specific Heat Capacity c

The **specific heat capacity** of a substance is the energy required to raise the temperature of 1 kg of that substance by 1 K (or 1 °C).

Energy and Temperature Change

$$Q = mc\Delta T \tag{2}$$

Q = energy supplied or removed (J)
m = mass (kg)
c = specific heat capacity ($\mathrm{J\,kg^{-1}\,K^{-1}}$)
ΔT = temperature change (K or °C)

✓ Exam Tip

In calorimetry experiments the energy supplied by an electric heater is $Q = IVt$. Setting $IVt = mc\Delta T$ and solving for c is a standard exam calculation.

2.4 Specific Latent Heat

During a **change of state** the temperature remains constant while energy is supplied or removed. This energy changes the *potential* energy of the molecules (breaking or forming bonds) without changing their kinetic energy.

Definition: Specific Latent Heat L

The **specific latent heat** of a substance is the energy required to change the state of 1 kg of the substance at constant temperature.

Latent Heat

$$Q = mL \tag{3}$$

Q = energy supplied or removed (J)
m = mass (kg)
L = specific latent heat ($\mathrm{J\,kg^{-1}}$)

Two types must be distinguished:

- **Specific latent heat of fusion** L_f — solid ↔ liquid
- **Specific latent heat of vaporisation** L_v — liquid ↔ gas

In general $L_v \gg L_f$ because during vaporisation molecules must be completely separated, requiring far more work against intermolecular forces than during melting.

! Common Mistake

Do not apply $Q = mc\Delta T$ when a substance is changing state. During a change of state the temperature does *not* change — use $Q = mL$ instead. A common error is forgetting to split a calculation into separate stages: warming, change of state, then further warming.

3 Topic 15 — Ideal Gases

3.1 The Mole and Avogadro's Constant

Definition: The Mole

One **mole** of a substance contains exactly $N_A = 6.02 \times 10^{23}$ particles, where N_A is the **Avogadro constant**.

Key relationships:

$$N = nN_A \qquad \text{(number of molecules from moles)} \tag{4}$$

$$n = \frac{m_{\text{sample}}}{M_r} \qquad \text{(moles from mass and molar mass)} \tag{5}$$

3.2 The Ideal Gas Equation

The empirical gas laws (Boyle's, Charles's, Pressure Law) combine into a single equation of state for an ideal gas:

Ideal Gas Equation

$$pV = nRT \tag{6}$$

$$pV = NkT \tag{7}$$

p = pressure (Pa)
V = volume (m^3)
n = amount of substance (mol)
R = molar gas constant = $8.31\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$
T = thermodynamic temperature (K)
N = number of molecules
k = Boltzmann constant = $1.38 \times 10^{-23}\,\text{J}\,\text{K}^{-1}$

The two constants are related by:

$$k = \frac{R}{N_A} \tag{8}$$

A gas **obeys** the ideal gas equation if $pV \propto T$ — equivalently, if pV/T is constant for a fixed mass of gas.

✓ Exam Tip

For a fixed mass of gas changing conditions:

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Use $pV = nRT$ when moles are given; use $pV = NkT$ when number of molecules is given.

3.3 Kinetic Theory of Gases

Assumptions of the Kinetic Model

- The gas consists of a **large number** of identical molecules.
- Molecules are in **continuous, random motion**.
- The **volume of the molecules** is negligible compared to the volume of the container.
- **Intermolecular forces** are negligible except during collisions.
- All collisions (molecule–molecule and molecule–wall) are **perfectly elastic**.
- The **duration of a collision** is negligible compared to the time between collisions.

Pressure from Kinetic Theory

Kinetic Theory Equation

$$pV = \tfrac{1}{3} N m \langle c^2 \rangle \tag{9}$$

N = total number of molecules
m = mass of one molecule (kg)
$\langle c^2 \rangle$ = mean-square speed ($\mathrm{m^2\,s^{-2}}$)

The **root-mean-square speed** is defined as:

$$c_{\text{r.m.s.}} = \sqrt{\langle c^2 \rangle} \tag{10}$$

Derivation Sketch

1. Consider one molecule of mass m, speed c_x normal to a wall of area A in a box of length L.
2. Momentum change per collision: $\Delta p = 2mc_x$.
3. Time between successive collisions with the same wall: $\Delta t = 2L/c_x$.
4. Force from one molecule: $F = mc_x^2/L$.
5. For N molecules, extend to 3-D using isotropy $\langle c_x^2 \rangle = \frac{1}{3}\langle c^2 \rangle$:

$$p = \frac{Nm\langle c^2 \rangle}{3V} \quad \Longrightarrow \quad pV = \tfrac{1}{3} N m \langle c^2 \rangle$$

3.4 Average Kinetic Energy of a Molecule

Comparing $pV = NkT$ with $pV = \frac{1}{3}Nm\langle c^2\rangle$:

Average Translational Kinetic Energy

$$\langle E_k\rangle = \tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT \tag{11}$$

The average kinetic energy of a molecule depends **only on temperature**.

! Common Mistake

$\langle E_k\rangle = \frac{3}{2}kT$ gives the kinetic energy of **one molecule**. For the total kinetic energy of n moles use $E_{\text{total}} = \frac{3}{2}nRT$. Do not confuse k (Boltzmann constant, per molecule) with R (molar gas constant, per mole).

4 Topic 16 — Thermodynamics

4.1 Internal Energy

Definition: Internal Energy U

The **internal energy** of a system is the sum of the random kinetic and potential energies of all its molecules. It is determined solely by the **state** of the system (its temperature, pressure, volume and composition).

For an **ideal gas** there are no intermolecular forces, so internal energy consists entirely of kinetic energy:

$$U = \tfrac{3}{2}NkT = \tfrac{3}{2}nRT \qquad \text{(ideal monatomic gas)} \tag{12}$$

Hence for an ideal gas, U depends *only* on T.

For **real substances** the potential energy between molecules is also significant. A rise in temperature increases kinetic energy; a change of state at constant temperature changes potential energy only.

4.2 Work Done by a Gas

Work Done at Constant Pressure

$$W = p\,\Delta V \tag{13}$$

W = work done **by** the gas (J)
p = pressure (Pa)
ΔV = change in volume (m^3)

Sign convention: expansion ($\Delta V > 0$) means the gas does positive work on the surroundings; compression ($\Delta V < 0$) means the surroundings do positive work on the gas.

4.3 The First Law of Thermodynamics

Definition: First Law of Thermodynamics

The increase in internal energy of a system equals the energy supplied to the system by heating *plus* the work done *on* the system:

$$\Delta U = q + W \tag{14}$$

Sign conventions (CIE syllabus form $\Delta U = q + W$):

Symbol	Positive when...	Negative when...
ΔU	internal energy increases	internal energy decreases
q	heat flows **into** system	heat flows **out** of system
W	work done **on** system	work done **by** system

✓ Exam Tip

CIE uses $\Delta U = q + W$ where W is work done **on** the gas. Some textbooks write $\Delta U = q - W$ where W is work done **by** the gas. Both are equivalent — always state your sign convention before substituting.

Special Processes

Process	Constraint	Consequence
Isothermal	$\Delta T = 0 \Rightarrow \Delta U = 0$	$q = -W$: heat in equals work out
Adiabatic	$q = 0$	$\Delta U = W$: compression raises U
Isochoric	$\Delta V = 0 \Rightarrow W = 0$	$\Delta U = q$: all heating changes U
Isobaric	$p = \text{const}$	$W = p\Delta V$; both q and W contribute

5 Formula Summary Sheet

Formula Summary Sheet

Formula	Meaning	Topic
$T/\mathrm{K} = \theta/°\mathrm{C} + 273.15$	Temperature conversion	14
$Q = mc\Delta T$	Heat and temperature change	14
$Q = mL$	Latent heat	14
$pV = nRT$	Ideal gas (molar form)	15
$pV = NkT$	Ideal gas (molecular form)	15
$k = R/N_A$	Boltzmann constant	15
$pV = \frac{1}{3}Nm\langle c^2\rangle$	Kinetic theory — pressure	15
$\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	Average KE per molecule	15
$c_{\mathrm{r.m.s.}} = \sqrt{\langle c^2\rangle}$	Root-mean-square speed	15
$U = \frac{3}{2}NkT = \frac{3}{2}nRT$	Internal energy (ideal gas)	16
$W = p\,\Delta V$	Work done by gas (const. p)	16
$\Delta U = q + W$	First law of thermodynamics	16

6 Exam Technique and Problem-Solving Strategy

6.1 5-Step Strategy for Thermal Physics Problems

1. **Identify the topic**: temperature/heat, ideal gas, or thermodynamics?
2. **List what is given** and what is being asked.
3. **Convert units**: temperatures to K; volumes to m^3; pressures to Pa.
4. **Select the correct formula** and rearrange *before* substituting.
5. **Check your answer**: correct units, sensible order of magnitude, correct sign.

6.2 Key Tips by Topic

Topic 14 — Temperature and Thermal Properties

- Split mixed heating problems into separate stages: warming + change of state + further warming.
- Remember $L_v \gg L_f$: for water, $L_v \approx 2.26 \times 10^6\,\mathrm{J\,kg^{-1}}$ vs $L_f \approx 3.34 \times 10^5\,\mathrm{J\,kg^{-1}}$.
- Calorimetry: set $IVt = mc\Delta T$; watch for heat losses.

Topic 15 — Ideal Gases and Kinetic Theory

- Use $pV = nRT$ for moles; use $pV = NkT$ for number of molecules.
- The kinetic theory derivation is examinable — practise the 3-step argument (momentum change → force → pressure, then extend to 3-D).
- R.m.s. speed $\propto \sqrt{T}$: doubling T in kelvin multiplies $c_{\text{r.m.s.}}$ by $\sqrt{2}$, *not* by 2.

Topic 16 — Thermodynamics

- Be rigorous about sign conventions: state them explicitly before substituting.
- For an isothermal process with an ideal gas, $\Delta U = 0$ always.
- Work done **on** the gas is $+W$ in CIE's first law convention.

7 Worked Examples

Question 1 An electric heater of power 60 W heats 0.50 kg of water from 20 °C to 70 °C in 175 s. Calculate the specific heat capacity of water.

Solution

Energy supplied: $Q = Pt = 60 \times 175 = 1.05 \times 10^4$ J
Using $Q = mc\Delta T$:

$$c = \frac{Q}{m\,\Delta T} = \frac{1.05 \times 10^4}{0.50 \times 50} = 4200\,\text{J kg}^{-1}\,\text{K}^{-1}$$

Question 2 A gas occupies $2.0 \times 10^{-3}\,\text{m}^3$ at a pressure of 1.5×10^5 Pa and temperature 27 °C. Calculate the number of moles of gas.

Solution

Convert: $T = 27 + 273.15 = 300.15 \approx 300\,\text{K}$

$$n = \frac{pV}{RT} = \frac{1.5 \times 10^5 \times 2.0 \times 10^{-3}}{8.31 \times 300} = \frac{300}{2493} = 0.120\,\text{mol}$$

Question 3 Calculate the r.m.s. speed of nitrogen molecules ($M_r = 28\,\text{g mol}^{-1}$) at 20 °C.

Solution

$$T = 293.15\,\text{K}; \quad m = \frac{28 \times 10^{-3}}{6.02 \times 10^{23}} = 4.65 \times 10^{-26}\,\text{kg}$$

$$\langle c^2 \rangle = \frac{3kT}{m} = \frac{3 \times 1.38 \times 10^{-23} \times 293.15}{4.65 \times 10^{-26}} = 2.61 \times 10^5\,\text{m}^2\,\text{s}^{-2}$$

$$c_{\text{r.m.s.}} = \sqrt{2.61 \times 10^5} = 511\,\text{m s}^{-1}$$

Question 4 A gas is compressed. The work done *on* the gas is 450 J and at the same time 200 J of heat is removed from the gas. Find the change in internal energy.

Solution

Work done on gas: $W = +450\,\text{J}$ (positive: on the gas)
Heat removed from gas: $q = -200\,\text{J}$ (negative: leaves system)

$$\Delta U = q + W = -200 + 450 = +250\,\text{J}$$

Internal energy *increases* by 250 J.

8 Practice Exam Questions

Section A — Short Answer Questions

Q1. Define specific heat capacity and state its SI unit. *[2 marks]*

Q2. Distinguish between the specific latent heat of fusion and the specific latent heat of vaporisation. *[2 marks]*

Q3. State *three* assumptions of the kinetic theory of gases. *[3 marks]*

Q4. A 2.0 kg block of ice at 0 °C is melted and the resulting water is heated to 100 °C. Given $L_f = 3.34 \times 10^5\,\mathrm{J\,kg^{-1}}$ and $c_{\text{water}} = 4200\,\mathrm{J\,kg^{-1}\,K^{-1}}$, calculate the total energy required. *[3 marks]*

Q5. The temperature of an ideal gas is doubled on the kelvin scale (at constant volume). State and explain what happens to:

- the r.m.s. speed of the molecules,
- the pressure of the gas.

[4 marks]

Section B — Longer Structured Questions

Q6. An ideal gas is enclosed in a container of fixed volume $5.0 \times 10^{-3}\,\mathrm{m}^3$. The gas contains 3.0×10^{23} molecules at a temperature of 300 K.

(a) Calculate the pressure of the gas. *[2 marks]*

(b) The temperature is raised to 450 K. Calculate the new pressure, assuming the gas remains ideal. *[2 marks]*

(c) Explain, using kinetic theory, why the pressure increases when the temperature is raised. *[3 marks]*

Q7. A sample of gas absorbs 600 J of heat from its surroundings and expands at a constant pressure of 1.2×10^5 Pa, increasing its volume by $2.0 \times 10^{-3}\,\mathrm{m}^3$.

(a) Calculate the work done *by* the gas. *[2 marks]*

(b) Using the first law of thermodynamics, calculate the change in internal energy of the gas. *[3 marks]*

(c) The gas is then compressed adiabatically back to its original volume. State what happens to the internal energy and explain why. *[2 marks]*

Q8. Show that the average translational kinetic energy of a molecule of an ideal gas is $\frac{3}{2}kT$. *[3 marks]*

9 Mark Scheme and Answers

Q1. Specific heat capacity is the energy required to raise the temperature of 1 kg of a substance by 1 K [1]; unit: $\mathrm{J\,kg^{-1}\,K^{-1}}$ [1].

Q2. Latent heat of *fusion*: energy per unit mass to change solid to liquid at constant temperature [1]. Latent heat of *vaporisation*: energy per unit mass to change liquid to gas at constant temperature [1].

Q3. Any three from: large number of molecules [1]; random motion [1]; volume of molecules negligible compared to container [1]; intermolecular forces negligible between collisions [1]; perfectly elastic collisions [1]; collision duration negligible [1]. (3 marks total)

Q4. Melting: $Q_1 = 2.0 \times 3.34 \times 10^5 = 6.68 \times 10^5\,\mathrm{J}$ [1]; heating water: $Q_2 = 2.0 \times 4200 \times 100 = 8.40 \times 10^5\,\mathrm{J}$ [1]; total: $Q = 1.508 \times 10^6\,\mathrm{J} \approx 1.51 \times 10^6\,\mathrm{J}$ [1].

Q5. R.m.s. speed: $c_{\text{r.m.s.}} \propto \sqrt{T}$, so doubling T increases $c_{\text{r.m.s.}}$ by factor $\sqrt{2} \approx 1.41$ [1]; because average KE increases [1]. Pressure: $p \propto T$ at constant V, so pressure doubles [1]; molecules collide with walls more frequently and with greater momentum [1].

Q6(a).

$$p = \frac{NkT}{V} = \frac{3.0 \times 10^{23} \times 1.38 \times 10^{-23} \times 300}{5.0 \times 10^{-3}} = 248\,\mathrm{Pa} \quad [2]$$

Q6(b). $p \propto T$ at constant V: $p_2 = 248 \times \dfrac{450}{300} = 372\,\mathrm{Pa}$ [2].

Q6(c). At higher temperature molecules move faster [1]; they collide with the walls more frequently [1]; each collision transfers greater momentum to the wall, so force and pressure increase [1].

Q7(a). $W_{\text{by gas}} = p\Delta V = 1.2 \times 10^5 \times 2.0 \times 10^{-3} = 240\,\mathrm{J}$ [2].

Q7(b). Work done *on* gas: $W = -240\,\mathrm{J}$ [1]; $\Delta U = q + W = 600 + (-240) = +360\,\mathrm{J}$ [1]; internal energy increases by 360 J [1].

Q7(c). Internal energy increases [1]; no heat enters or leaves (adiabatic), so all work done on the gas during compression increases internal energy [1].

Q8. From kinetic theory: $pV = \frac{1}{3}Nm\langle c^2\rangle$ [1]; from ideal gas: $pV = NkT$; equating and simplifying: $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ [1]; this equals the average translational kinetic energy of one molecule [1].

10 Revision Checklist

Revision Checklist

Learning Objective	Confidence (1–3)
Topic 14 — Temperature	
☐ Explain thermal equilibrium and state the zeroth law	
☐ Convert temperatures between °C and K using $T = \theta + 273.15$	
☐ State and explain examples of thermometric properties	
☐ Define and use specific heat capacity: $Q = mc\Delta T$	
☐ Define specific latent heat; distinguish L_f and L_v	
☐ Apply $Q = mL$; split calculations across stages correctly	
Topic 15 — Ideal Gases	
☐ Use molar quantities and the Avogadro constant N_A	
☐ Recall and apply $pV = nRT$ and $pV = NkT$	
☐ State the relationship $k = R/N_A$	
☐ State all six assumptions of the kinetic theory	
☐ Reproduce the derivation of $pV = \frac{1}{3}Nm\langle c^2\rangle$	
☐ Derive the result $\langle E_k\rangle = \frac{3}{2}kT$	
☐ Calculate r.m.s. speed from temperature and molar mass	
Topic 16 — Thermodynamics	
☐ Explain internal energy in terms of molecular KE and PE	
☐ State that internal energy of an ideal gas depends only on T	
☐ Calculate work done by/on a gas: $W = p\Delta V$	
☐ Apply the first law $\Delta U = q + W$ with correct sign conventions	
☐ Analyse isothermal, adiabatic, isochoric and isobaric processes	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Remember: understanding why *formulae work is more powerful than memorising them.*

Practise deriving $pV = \frac{1}{3}Nm\langle c^2\rangle$ *and* $\langle E_k\rangle = \frac{3}{2}kT$ *from first principles.*